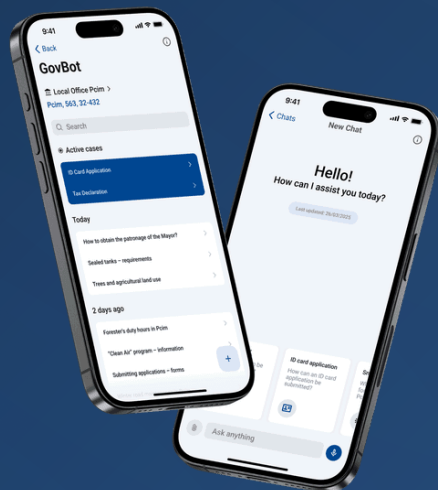


1ST PLACE -- MINISTRY OF DIGITAL AFFAIRS CHALLENGE

GovBot

AI Assistant for Local Governments in mObywatel -- built in 24 hours at the Econverse Hackathon



Event	Track	Duration	Date
Econverse Hackathon	Smart Cities	24 hours	26-27 March 2025

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1. Overview

GovBot is an AI-powered conversational assistant designed to be integrated into **mObywatel** -- the official Polish government mobile app with over **10 million active users**. It provides citizens with instant, 24/7 access to local government information through both text and voice interactions.

1st Place -- Econverse Hackathon

Ministry of Digital Affairs Challenge | Smart Cities Track



Econverse Hackathon -- 26-27 March 2025, Warsaw & Lodz, Poland



The team presenting GovBot on stage at the Econverse Hackathon

2. The Challenge

The project was created for the challenge proposed by the **Polish Ministry of Digital Affairs**, part of the E converse Hackathon's Smart Cities track -- focused on building solutions for the cities of the future.

Design an AI-powered assistant that provides citizens with quick, intuitive access to local government services and information.



Ministry of Digital Affairs
Republic of Poland

Ministry of Digital Affairs, Republic of Poland -- Challenge Sponsor

Local governments across Poland are struggling with **bureaucracy and information overload**. The challenge focused on creating a user-friendly digital solution that makes local services more accessible, helping residents quickly find the information they need. The outcome needed to be simple, intuitive, and easy for local governments to implement.



Teams working during the 24-hour hackathon at Kozminski University

3. Problem Statement

How might we help citizens find relevant local government information faster and more intuitively, while reducing the workload of municipal staff?

Pain Points -- Citizen's Perspective

Scattered information

Difficulty finding relevant details across multiple websites

Unclear deadlines & requirements

Important dates and needed documents often buried or outdated

Long waiting times

Both online (slow email responses) and offline (queues at offices)

Bureaucratic language

Information written in legal jargon, hard to understand

Pain Points -- Institution's Perspective

Repetitive inquiries

Answering the same simple questions every day

Work overload

Too many calls, emails, and visits that could be handled automatically

Fragmented systems

Lack of integration between databases and communication channels

Limited resources

Not enough staff to provide quick and personalized support

4. Research & Analysis

Desk Research

The **NIK report** "Provision of Public Services" found that in approximately **30% of surveyed offices**, information about e-services was unclear and difficult to find on their websites.

Reports on the digitalization of public offices indicate that many inquiries could be handled automatically by IT systems, but are not -- resulting in people calling, writing, or visiting in person.

Competitive Analysis

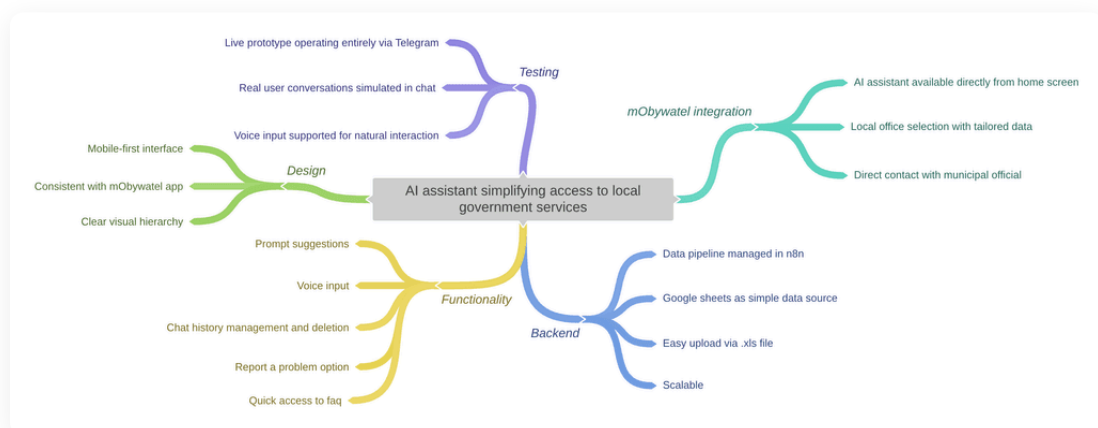
Comparison of channels citizens use to access local government information, based on speed and level of staff involvement:



Competitive analysis: GovBot positioned as the fastest access channel with the lowest resource requirements

Ideation -- Mindmap

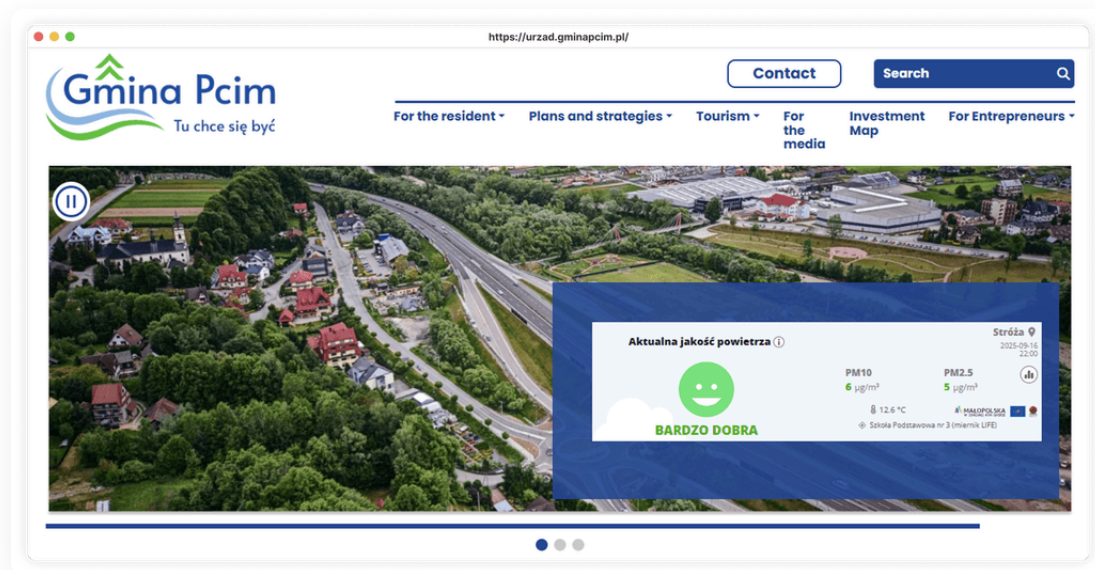
The team brainstormed ideas focused on providing citizens with intuitive access to local government services through a widely used platform.



Mind map exploring core areas: mObywatel integration, design, functionality, backend, and testing

Real-World Data Source

For the research and prototyping phase, the team focused on the real-life example of the **Pcim municipality** to gather actual local government data and test the solution in a specific context.



Official Pcim municipality website -- the data source used for the GovBot prototype

5. Solution

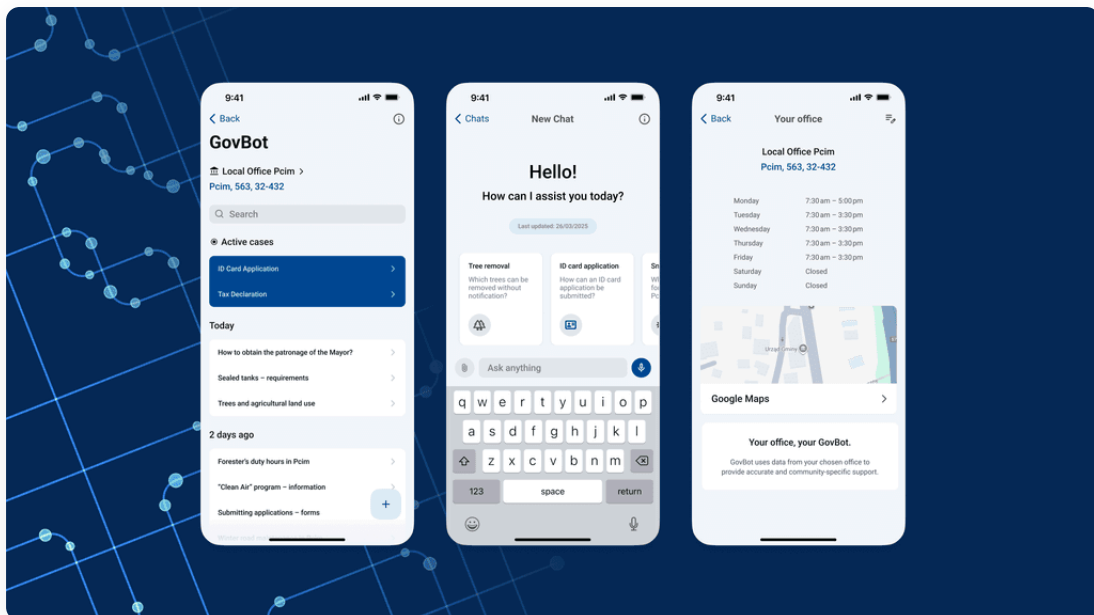
GovBot is an AI assistant integrated into the Polish national app mObywatel, providing citizens with instant access to local government information.

Value Proposition

By introducing GovBot to mObywatel, citizens gain faster and more intuitive access to local government information, while municipal staff experience a reduced workload and fewer repetitive inquiries.

Core Benefits

- **Easy data management** -- Municipalities can update data by uploading a .XLS file
- **24/7 access** -- Citizens get quick and intuitive access to local government info around the clock
- **Massive reach** -- Integrated into mObywatel, reaching over 10 million users
- **Scalable** -- Deployable to multiple municipalities across Poland
- **Efficiency** -- Reduces workload for municipal offices



GovBot's core screens: chat history, new conversation, and local office selection

6. Key Features

Feature 1: mObywatel Integration

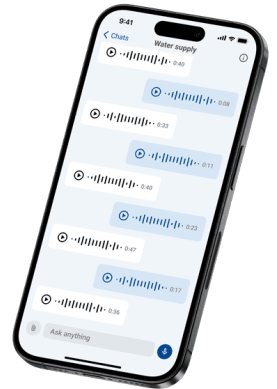
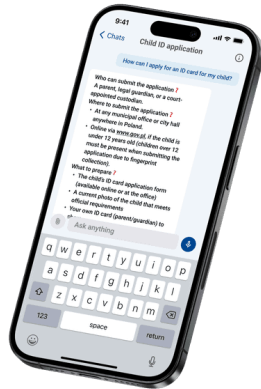
- Reaches over 10 million active users in Poland
- Built on a trusted, government-backed platform
- Eliminates the need for residents to install a new app
- Makes scaling across municipalities fast and cost-effective

Feature 2: Prompt Suggestions

- Quick start without worrying about wording
- Covers common topics: documents, deadlines, fees, events
- Adapts to user's context and history

Feature 3: Conversational Interaction

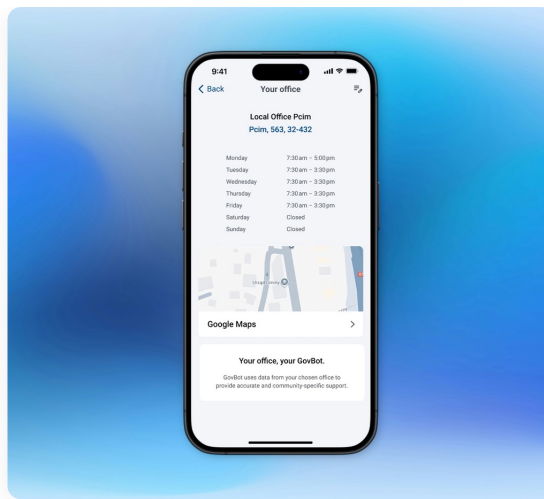
- Ask in your own words -- no need to know official terms
- Feels like chatting in any popular messenger
- No need to browse menus or search across multiple pages



GovBot supports both text-based (left) and voice-based (right) conversations

Feature 4: Local Office Selection

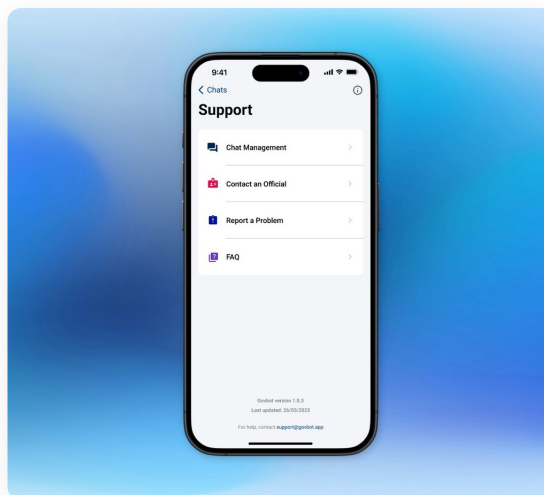
- Delivers tailored information from the user's specific local office
- Lets users switch offices if they want info from another location
- Office details at a glance: address, hours, contacts



Local office selection screen with address, opening hours, and map

Feature 5: Support & Help

- Manage past chats to keep conversations organized
- Contact local officials directly
- Report any issues or problems
- Access FAQs

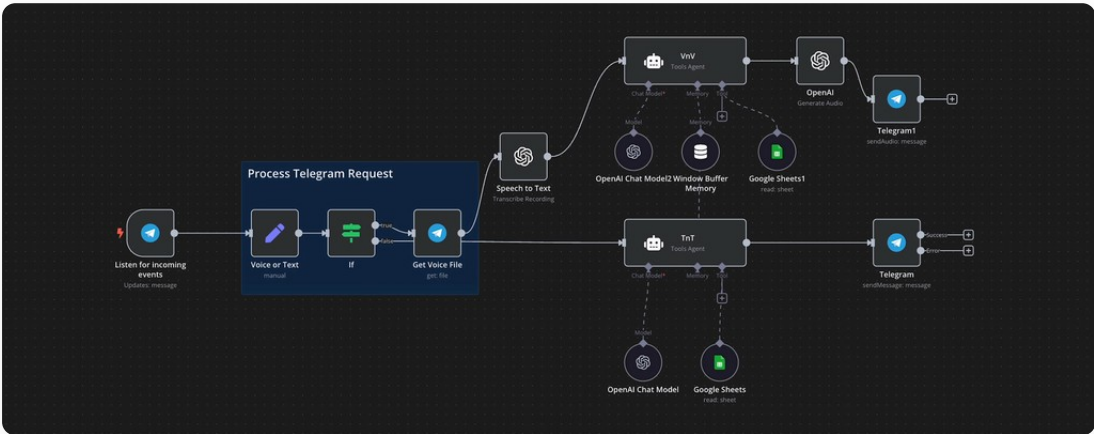


Support menu with chat management, contact, problem reporting, and FAQ access

7. Architecture & Tech Stack

The conversational AI model was built entirely in **n8n**, allowing easy iteration for both text-based and voice interactions. The team scraped content from the official Pcim municipality website and organized it into a structured knowledge base using Google Sheets.

The assistant was configured with a **low temperature setting** to avoid hallucinations and strictly referenced the connected knowledge source.



Complete n8n automation workflow showing both text-to-text and voice-to-voice pipelines

Text-to-Text Pipeline

Component	Technology
Language Model	ChatGPT 4.5 Preview
Data Source	Local DB Access
Q&A System	Dedicated Q&A System

Voice-to-Voice Pipeline

Component	Technology
Language Model	ChatGPT 4o mini
Speech-to-Text	Whisper Transcription
Text-to-Speech	TTS-1 Speech
Logic Layer	Input-Based Logic
Multilingual	Language Detection

Knowledge Base

Knowledge [Pcim]					
File Edit View Insert Format Data Tools Extensions Help					
Q Menu					
K20					
1	A	B	C	D	E
2	Source	Sample Question	Chatbot Knowledge	Single String Format (Q+A)	Handover
1	Pages ...		Program „Czyste Powietrze” ma na celu poprawę jakości powietrza poprzez wsparcie finansowe dla osób wymieniających stare kotły i piece na nowoczesne urządzenia spełniające wymagania ekoprojektu. W Gminie Pćim dostępne są punkty konsultacyjne, które pomagają w wypełnianiu wniosków.	Jakie są najważniejsze informacje dotyczące Programu „Czyste Powietrze”?	(No Handover)
2	Pages ...	Jakie są najważniejsze informacje dotyczące Programu „Czyste Powietrze”?	Na dzień 30.09.2024 r. złożono 581 wniosków, zawarto 397 umów, zrealizowano 213 przedsięwzięć, a sumaryczna kwota wypłaconych dotacji wyniosła 5 493 444,56 zł.	Jakie są statystyki dotyczące liczby wniosków i wypłaconych dotacji w ramach Programu „Czyste Powietrze”?	(No Handover)
3	Pages ...	Jakie są statystyki dotyczące liczby złożonych wniosków i wypłaconych dotacji w ramach Programu „Czyste Powietrze”?	Na terenie Gminy Pćim działają trzy apteki.	Ile aptek znajduje się na terenie Gminy Pćim?	(Suggested)
4	Pages ...	Ile aptek znajduje się na terenie Gminy Pćim?	Od 1 lipca 2017 roku obowiązuje zakaz montażu nowych kotłów na węgiel lub drewno o parametrach emisji gorszych niż wyznaczone w unijnych rozporządzeniach. Do końca kwietnia 2024 roku konieczna będzie wymiana kotłów, które nie spełniają żadnych norm emisyjnych. Od 1 maja 2024 roku dopuszczone będzie używanie tylko kominków spełniających wymagania ekoprojektu.	Jakie są główne ograniczenia dotyczące eksploatacji kotłów i kominków na paliwa stałe w Małopolsce?	(No Handover)
5	Pages ...	Jakie są główne ograniczenia dotyczące eksploatacji kotłów i kominków na paliwa stałe w Małopolsce?	Krzewy rosnące w skupisku o powierzchni do 25 m² oraz krzewy na terenach pokrytych roślinnością pełniącą funkcję ochronną, z wyłączeniem krzewów w pasie drogowym drogi publicznej, na terenie nieruchomości wpisanej do rejestru zabytków oraz na terenach zieleni.	Jakie krzewy można usunąć bez zgłoszenia?	(No Handover)
6	Pages ...	Jakie krzewy można usunąć bez zgłoszenia?	Wnioski dotyczące wyřbu drzew można pobierać ze strony internetowej nadleśnictwa w zakładce: „Nasza praca → Nadzór nad lasami niepaństwowymi” oraz składać do biura Nadleśnictwa Myślenice za pośrednictwem poczty tradycyjnej (adres: ul. Szpitalna 13, 32-400 Myślenice) lub elektronicznie na adres: myslenice@krakow.lasy.gov.pl.	Jak można składać wnioski dotyczące wyřbu drzew?	(Suggested)
7	Pages ...	Jak można składać wnioski dotyczące wyřbu drzew?	Zezwolenie wyřasa w przypadku: 1) likwidacji punktu sprzedaży; 2) upływu terminu ważności zezwolenia; 3) zmiany rodzaju działalności punktu sprzedaży; 4) zmiany składu osobowego wspólników spółki cywilnej; 5) niedopełnienia obowiązku złożenia oświadczenia lub dokonania opłaty.	Kiedy zezwolenie na sprzedaż alkoholu wyřasa?	(No Handover)
8	Pages ...	Kiedy zezwolenie na sprzedaż alkoholu wyřasa?	Stawki podatków lokalnych na rok 2025 obejmują: Podatek od nieruchomości, Podatek rolny, Podatek leśny, Podatek od środków transportowych oraz zwrot podatku akcyzowego zawartego w cenie oleju nadeńowego.	Jakie są stawki podatków lokalnych?	(No Handover)
9	Pages ...				

Structured knowledge base in Google Sheets with municipal Q&A data from Pćim

Tools Used

Category	Tools
Design	Figma, Adobe Photoshop
Video & Motion	Adobe Premiere Pro, Adobe After Effects
Automation	n8n
Prototyping	Telegram
Data	Google Sheets

Key Insight: By enabling the assistant through a dedicated Telegram channel, testing became highly accessible -- every hackathon participant could engage with the prototype directly on their phone.

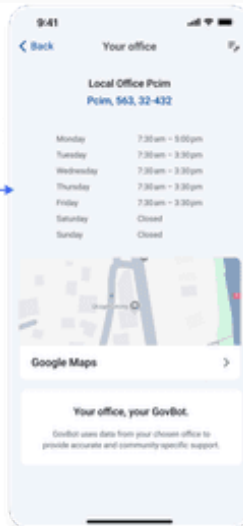
8. Design Process

The project followed a structured UX design process within the 24-hour hackathon constraint: Desk Research, Competitive Analysis, Service Blueprint, Pain Points identification, Goals definition, Mindmap brainstorming, Value Proposition, UI Reverse Engineering, Design, and Build.

UI Reverse Engineering

One of the main challenges in implementing GovBot into mObywatel was the **lack of a publicly available design system**. The existing design patterns and components of the mObywatel app were reverse-engineered to ensure seamless integration. GovBot's interface was built based on the guidelines and patterns identified in the mObywatel app.

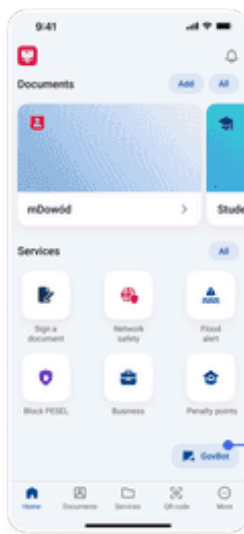
Complete UI Design



Detailed information about the nearest municipal office from which GovBot retrieves data, with the option for users to select a different office if needed.

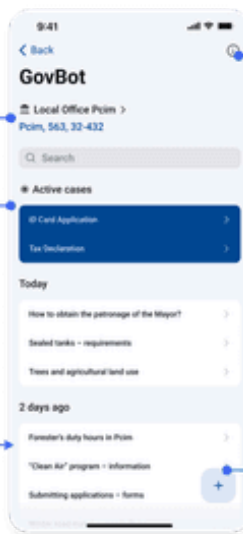


A list of suggested queries that helps users interact with GovBot efficiently, reducing friction for first-time interactions.

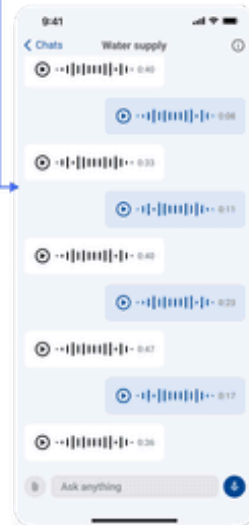


A new entry point to GovBot was added in the bottom navigation, allowing users to access the assistant quickly and intuitively, without the need to explore deeper parts of the interface.

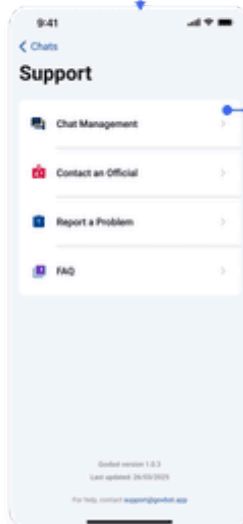
Ongoing matters and processes that the user is currently involved in.



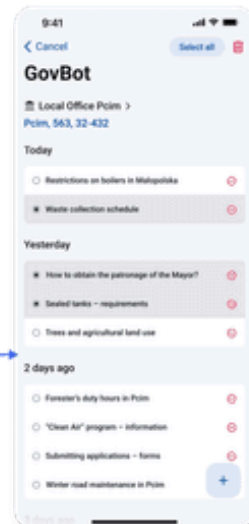
Overview of all chat interactions with GovBot. Users can quickly browse recent conversations, access past chats, and use the search function to find specific topics.



Voice conversation using audio messages, giving users the ability to send questions and later access audio answers.



Users can manage chats, contact an official directly, or



Select one or multiple chats to delete, giving users control

report an issue.

over their conversation history.

Complete set of GovBot UI screens designed for integration with mObywater

9. Team

Wiktor Lis

UX/UI Designer
Project Lead
Pitch Presenter

Katarzyna Stepień

AI Lead

Piotr Szewczyk

Information Architect

Jakub Pydzik

Data Engineer



The team presenting GovBot at the Econverse Hackathon stage

10. Key Takeaways

Lesson 1: Fitting the pieces without a guide

Integrating GovBot into mObywatel showed the challenge of reverse-engineering a design system and how essential fidelity is for realistic prototypes.

Lesson 2: Prioritize and iterate fast

In a hackathon, time is limited. Focusing on key features and iterating fast let the team deliver a functional solution within 24 hours.

Lesson 3: Design with the real world in mind

Working with a real municipality and real user data showed that knowing existing workflows from the start helps create usable solutions.

11. References

- Project showcase: wiktoria.framer.website/projects/govbot
- NIK Report -- Provision of Public Services: nik.gov.pl
- Digitalization Report: geoforum.pl

GovBot -- AI Assistant for Local Governments in mObywatel

Built in 24 hours at the E converse Hackathon | March 2025